

Trilytics Challenge

Build a Statistical Model and Framework to Generate District Level Market Attractiveness Index (Indian Pharmaceutical Industry) scores for over 700+ districts in India

Executive Summary

Sun Pharma aims to build a statistical model and framework to generate 'District Level Market (Pharmaceutical) Attractiveness Index' scores for over 700+ districts in India.

The objective is to move beyond traditional market assessment approaches and develop a comprehensive, data driven and forward-looking framework that captures the underlying drivers of pharmaceutical opportunity at a granular geographic level. The expected solution would be a **structured analytical framework that integrates multiple input variables to generate district level Market Attractiveness Index scores as its key output**. Participants are expected **use all available public domain and secondary data** to design a model that generates three distinct Index scores for each district:

- **Overall Market Attractiveness Index**
- **Chronic Therapy Market Attractiveness Index**
- **Acute Therapy Market Attractiveness Index**

The framework should integrate multiple dimensions such as macroeconomic factors, disease burden, patient potential, infrastructure, provider availability, affordability, healthcare spending and leading indicators of future demand. It should also provide a forward-looking perspective on how district level opportunity is expected to evolve.

This framework will enable more effective **sales force deployment & expansion, market prioritization and resource allocation** in the pharmaceutical industry.

Background

In the pharmaceutical industry, market opportunities vary significantly across districts due to differences in economic profiles, disease burden, patient pools, infrastructure, access, affordability, and growth potential.

However, current approaches to market planning and sales force deployment are often **limited in scope**. Decisions are typically driven by factors such as:

- Competition benchmarking
- Doctor Universe and coverage
- Internal indicators

These approaches do not fully capture the **true underlying market potential**.

Key gaps in current practice include:

- Limited incorporation of **disease prevalence and patient pool**
- Lack of integration of **healthcare spending and affordability indicators**
- Inadequate consideration of **access gaps (e.g., patient-to-doctor ratio)**
- Absence of **forward-looking insights** on how markets will evolve
- Minimal use of **leading indicators** such as income growth, urbanisation, and risk factor trends
- Lack of a unified, comparable **district-level framework**

As a result, current approaches may overlook:

- Emerging high growth districts
- Underserved markets with strong latent demand
- Future shifts in therapy demand

This creates a need for a **holistic, data-driven and predictive framework that can better inform strategic decision-making**.

Objectives

Design and build a **district level Market Attractiveness Index Framework and statistical model for the pharmaceutical industry** that systematically evaluates and ranks all districts in India based on their strategic business potential.

The framework should translate diverse market drivers into **quantifiable Market Attractiveness Index scores for each district**, enabling clear comparison and prioritization across geographies.

Your solution should:

- Generate **three separate district level indices**:
 - **Overall Market Attractiveness Index**
 - **Chronic Therapy Market Attractiveness Index**
 - **Acute Therapy Market Attractiveness Index**
- Incorporate a **comprehensive set of variables**, including but not limited to:
 - Macroeconomic indicators
 - Disease prevalence and epidemiology
 - Patient pool and patient-to-doctor ratio
 - Provider availability and doctor density
 - Healthcare infrastructure and access
 - Healthcare spending and affordability
 - Leading indicators for future demand (e.g., income growth, urbanization, risk factors)
 - Any other relevant factors with proper explanation and justification
- Provide:
 - Current state view of market attractiveness
 - Forward looking view of future opportunity at the district level

Market Attractiveness Index – Definition

The **Market Attractiveness Index** is a **composite scoring mechanism** that assigns a score to each district based on a combination of relevant variables.

- Each district will receive a **numerical score representing its relative market attractiveness**
- **Higher scores indicate higher strategic business potential**
- The index should enable **comparison and ranking across districts**



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Participants are responsible for:

- Selecting the variables included in the index
- Defining the relevance and impact of each variable
- Designing the scoring and weighting methodology
- Development of any models (ML/DL/GenAI etc.) as relevant to the case

Data Guidelines

Participants are expected to source and integrate **public domain and other relevant datasets** to build the framework. Relevant sources may include:

- Central, State and district level data from Government portals
- Census data
- WHO, World Bank, ICMR, NFHS
- Research publications and open datasets
- RK SWAMY or other credible secondary sources
- Any other relevant datasets that strengthen the analysis

Participants should clearly document:

- Data sources used
- Variable definitions
- Assumptions and proxies
- Rationale for data selection

What You Need to Deliver

1. District Level Market Attractiveness Framework and Index Output

Develop a **district level Market Attractiveness Framework** that generates and ranks **Market Attractiveness Index scores for all districts** across:

- **Overall pharmaceutical market attractiveness index**
- **Chronic therapy market attractiveness index**
- **Acute therapy market attractiveness index**



Clearly explain:

- What each index represents
- How they differ
- How they can be used for decision making

2. Variable Selection and Business Rationale

Explain:

- Why each variable is included
- How it contributes to pharmaceutical market attractiveness
- How variables differ across acute and chronic segments
- How your approach improves upon current industry practices

3. Methodology

Provide detailed explanation of:

- Data preparation and transformations
- Normalization and weighting
- Index construction
- Handling missing data
- Incorporation of **future demand indicators**
- Approach for **predicting future district level opportunity**

4. Business Application

Demonstrate how your framework supports:

- **Sales force deployment & expansion**
- Market prioritization
- Territory planning

Show how it helps identify:

- High potential districts today
- **Future high growth districts**



5. Visualization and Insights

Present outputs using clear visualizations such as:

- District rankings
- Maps and heatmaps
- Therapy wise comparisons
- Current vs future opportunity views

6. Replicability and Scalability

Explain:

- how the framework can be reused across therapies
- adaptability for acute vs chronic segments
- scalability across geographies

IMPORTANT:

- All final submissions must be uploaded as a single ZIP file containing the complete set of relevant submission materials.
- Each ZIP file must include **one core PowerPoint presentation** that presents and explains the submission in detail. This PowerPoint will be considered the primary document for evaluation.
- **Participants are welcome to include any additional supporting files, beyond the core PowerPoint presentation, that may help support or strengthen their case submission.**

Evaluation Criteria

Submissions will be evaluated holistically across the following dimensions:

Dimension	What Will be Evaluated
Data, Variables & Assumptions	Quality and relevance of data sources used, selection of variables, use of proxy logic where required, treatment of data quality issues, clarity of assumptions, transformations applied, handling of missing data, and relevance of the chosen inputs to pharma market attractiveness.
MAI Framework, Model Design & Validation	Design of the overall Market Attractiveness Index, Chronic MAI and Acute MAI; logic behind the framework; normalization approach; model structure; statistical or analytical rigor; explainability of the methodology; validation approach or accuracy checks; and scalability of the model.
Insights, Business Actionability & Implementation	Quality of insights derived from the analysis, district rankings, current versus future opportunity assessment, relevance for sales-force deployment, market prioritization, territory planning, practicality of recommendations, and clarity of the implementation roadmap.
PPT Communication, Visualization & Innovation	Clarity of the presentation storyline, quality of executive-level communication, effective use of visuals such as maps, heatmaps, charts or rankings, and the ability to present insights in a clear and compelling way. Candidates may also be recognized for using any innovative methods, or creative approaches that strengthen the solution.



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IMPORTANT NOTES:

1. All queries related to the case, in any form, may be posted through the following link: <https://forms.gle/xPJboeXx58ZAmmp9A>
2. Please note that responses may take a few days, depending on the availability of the business representatives and the volume of queries received.
3. **A separate Ask Me Anything (AMA) session is tentatively scheduled for 08.07.2026 at 19:00 IST.** Participants are strongly encouraged to attend this session, where a Senior Manager from Sun Pharma will explain the case and share key dos and don'ts.
4. The AMA Session meeting link will be shared one day before to your registered email id. Keep an eye on the email and spam folders. Contact the Trilytics team in case of any issues.
5. Participants are advised to go through this document thoroughly and begin working on their case submission in advance. This will help them consolidate their questions and get them clarified during the AMA session, thereby enabling a more effective and well-informed submission.

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